Exp - 6.1

→ Write a c program that uses functions to perform the following operations
a) Reading a complex number
b) Writing a complex nubmer
c) Addition and subtraction of two complex number

```
# include < stdio . h >

// structure to represent a complex number
struct complex {
      float real ;
      float imag ;
} ;

// Function to read a complex number
struct complex readcomplex () {
  struct complex num ;
  printf (" Enter the real part :  " );
  scanf (" %f " , & num. real );
  printf (" Enter the imaginary part : );
  scanf (" %f ", & num. imag );
  return num ;
}
```

```
// Function to display a complex number
void writeComplex(struct Complex num) {
    printf("%.lf + %.lfi", num.real, num.imag);
}

// Function to add two complex numbers
struct Complex addComplex(struct complex n1, struct
complex n2) {
    struct Complex result;
    result.real = n1.real + n2.real;
    result.imag = n1.imag + n2.imag;
    return result;
}

int main()
{
    struct Complex num1, num 2, sum, difference;

printf("Enter the first complex number is:\n");
    num1 = readComplex();

printf("\n Enter the second complex number:\n");
    num 2 = readComplex();

sum = addComplex(num1, num 2);
difference = subtractComplex(num1, num 2);
```

```
printf ("\n The first complex number is : ");
 WriteComplex (num1) ;

printf ("\n the second complex number is : ");
 writeComplex (num 2);

printf ("\n\n The sum of the two complex number is : ");
 writeComplex ( sum);

printf ("\n The difference of the two complex numbers is:
 writeComplex (difference );

return 0 ;

}
```

```c
#include <stdio.h>

// Structure to represent a complex number
struct Complex {
    float real;
    float imag;
};

// Function to read a complex number
struct Complex readComplex() {
    struct Complex num;
    printf("Enter the real part: ");
    scanf("%f", &num.real);
    printf("Enter the imaginary part: ");
    scanf("%f", &num.imag);
    return num;
}

// Function to display a complex number
void writeComplex(struct Complex num) {
    printf("%.1f + %.1fi", num.real, num.imag);
}

// Function to add two complex numbers
struct Complex addComplex(struct Complex n1, struct Complex n2) {
    struct Complex result;
    result.real = n1.real + n2.real;
    result.imag = n1.imag + n2.imag;
    return result;
}

// Function to subtract two complex numbers
struct Complex subtractComplex(struct Complex n1, struct Complex n2) {
    struct Complex result;
    result.real = n1.real - n2.real;
    result.imag = n1.imag - n2.imag;
    return result;
}

int main() {
    struct Complex num1, num2, sum, difference;

    printf("Enter the first complex number:\n");
    num1 = readComplex();

    printf("\nEnter the second complex number:\n");
    num2 = readComplex();

    sum = addComplex(num1, num2);
    difference = subtractComplex(num1, num2);

    printf("\nThe first complex number is: ");
    writeComplex(num1);

    printf("\nThe second complex number is: ");
    writeComplex(num2);

    printf("\n\nThe sum of the two complex numbers is: ");
    writeComplex(sum);

    printf("\nThe difference of the two complex numbers is: ");
    writeComplex(difference);

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output> & .\'exp1.exe'
Enter the first complex number:
Enter the real part: 4
Enter the imaginary part: 3

Enter the second complex number:
Enter the real part: 1
Enter the imaginary part: 2

The first complex number is: 4.0 + 3.0i
The second complex number is: 1.0 + 2.0i

The sum of the two complex numbers is: 5.0 + 5.0i
The difference of the two complex numbers is: 3.0 + 1.0i
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>
```

Exp 6.2

→ Write a C program to compute the monthly pay of 100 employees using each employee-s name, basic pay. The DA computed as 52% of the basic pay Gross - salary (basic pay + DA). Print the employees name and gross salary

```
# include < stdio.h >

struct Employee {
    char name [50] ;
    float basicPay ;
    float da ;
    float grossSalary ;
  } ;

int main () {
    struct Employee emp [100] ;
    int i, n;

printf (" Enter the number of employees (up to 100): ");
scanf ( "%d", &n);

for (i = 0 ; i < n ; i++) {
printf ("\n Enter details of employee %d :\n", i+1);
```

```
printf ("Enter name: ");
scanf ("%s", emp[i].name);
printf ("Enter basic pay: ");
scanf ("% f", &emp[i].basicPay);

emp[i].da = 0.52 * emp[i].basic Pay ;
emp[i].grossSalary = emp[i].basicPay + emp[i].da;
}

printf ("\n Employee Name and gross salary details:\n");
printf ("-----------------------\n");

for (i=0 ; i<n ; i++)
 {
 printf ( " Employee Name: %s \n", emp[i].name);
 printf (" gross salary: %.2f \n ", emp[i].grossSalary);
 printf ("-------------------\n");
 }

return 0;
}
```

```c
#include <stdio.h>

struct Employee {
    char name[50];
    float basicPay;
    float da;
    float grossSalary;
};

int main() {
    struct Employee emp[100];
    int i, n;

    printf("Enter the number of employees (up to 100): ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("\nEnter details of employee %d:\n", i + 1);
        printf("Enter name: ");
        scanf("%s", emp[i].name);
        printf("Enter basic pay: ");
        scanf("%f", &emp[i].basicPay);

        emp[i].da = 0.52 * emp[i].basicPay;
        emp[i].grossSalary = emp[i].basicPay + emp[i].da;
    }

    printf("\nEmployee Name and Gross Salary Details:\n");
    printf("--------------------------------------\n");

    for (i = 0; i < n; i++) {
        printf("Employee Name: %s\n", emp[i].name);
        printf("Gross Salary: %.2f\n", emp[i].grossSalary);
        printf("--------------------------------------\n");
    }

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output> & .\'exp2.exe'
Enter the number of employees (up to 100): 5

Enter details of employee 1:
Enter name: sanchita
Enter basic pay: 20000

Enter details of employee 2:
Enter name: aanya
Enter basic pay: 21390

Enter details of employee 3:
Enter name: aryan
Enter basic pay: 19028

Enter details of employee 4:
Enter name: bhumi
Enter basic pay: 30000

Enter details of employee 5:
Enter name: sehajdeep
Enter basic pay: 18565

Employee Name and Gross Salary Details:
--------------------------------------
Employee Name: sanchita
Gross Salary: 30400.00
--------------------------------------
Employee Name: aanya
Gross Salary: 32512.80
--------------------------------------
Employee Name: aryan
Gross Salary: 28922.56
--------------------------------------
Employee Name: bhumi
Gross Salary: 45600.00
--------------------------------------
Employee Name: sehajdeep
Gross Salary: 28218.80
--------------------------------------
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>
```

Exp 6.3

Create a Book structure containing book_id, title, author name & price. Write a C program to pass a structure as a function argument and print the book details

→

```
#include < stdio.h >

struct Book {
 int book_id;
 char title [100];
 char author [100];
  float price;
};

void displayBook ( struct Book b) {
 printf ("\n Book Details : \n");
 printf ("Book ID : %d \n" , b.book_id);
 printf (" Title : %s \n", b.title);
 printf (" Author : %s \n", b.author);
 printf (" Price : %.2f \n", b.price);
}

int main () {
 struct Book b1 ;
```

EXPT. NO. NAME Date: 23/10

```
printf("Enter Book ID : ");
scanf("%d", &b1.book_id);

printf("Enter Book Title : ");
scanf(" %[^\n]", b1.title);

printf("Enter Author Name: ");
scanf(" %[^\n]", b1.author);

printf("Enter Book Price: ");
scanf("%f", &b1.price);

displayBook(b1);

return 0;
}
```

```c
#include <stdio.h>
#include <stdlib.h>

struct Book {
    int book_id;
    char title[100];
    char author[100];
    float price;
};

void displayBook(struct Book b) {
    printf("\nBook Details:\n");
    printf("Book ID: %d\n", b.book_id);
    printf("Title: %s\n", b.title);
    printf("Author: %s\n", b.author);
    printf("Price: %.2f\n", b.price);
}

int main() {
    struct Book b1;

    printf("Enter Book ID: ");
    while (scanf("%d", &b1.book_id) != 1) {
        printf("Invalid input! Please enter a numeric Book ID: ");
        while (getchar() != '\n');
    }

    printf("Enter Book Title: ");
    scanf(" %[^\n]", b1.title);

    printf("Enter Author Name: ");
    scanf(" %[^\n]", b1.author);

    printf("Enter Book Price: ");
    while (scanf("%f", &b1.price) != 1) {
        printf("Invalid input! Please enter a valid price: ");
        while (getchar() != '\n');
    }

    displayBook(b1);
    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output> & .\'exp3.exe'
Enter Book ID: 101
Enter Book Title: C Programming
Enter Author Name: Dennis Ritchie
Enter Book Price: 450

Book Details:
Book ID: 101
Title: C Programming
Author: Dennis Ritchie
Price: 450.00
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>
```

6.4

→

Create a union containing 6 strings: name, home address, hostel_address, city, state, zip. Write a C program to display your present address

```
# include < stdio.h >
# include < string.h >

union Address {
  char name [50];
  char home-address [100];
  char hostel_address [100];
  char city [50];
  char state [50];
  char zip [10];
};

int main () {
  union address a;

printf (" Enter your present address: \n");

printf (" Name: ");
scanf (" %[^\n]", a.name);
```

Teacher's Signature:

Date: 29/10/25

```
printf("Home Address : ");
scanf(" %[^\n]", a.city);

printf(" city : ");
scanf(" %[^\n]", a.city);

printf(" State : ");
scanf(" %[^\n]", a.state);

printf( "ZIP code : ");
scanf(" %[^\n]", a.zip);

printf("\n Your Present Address is : \n");
printf("%s\n" , a.name);
printf("%s\n" , a.home_address);
printf("%s, %s - %s \n", a.city, a.state, a.zip);

return 0;

}
```

Teacher's Signature:

```c
#include <stdio.h>
#include <string.h>

union Address {
    char name[50];
    char home_address[100];
    char hostel_address[100];
    char city[50];
    char state[50];
    char zip[10];
};

int main() {
    union Address a;

    printf("Enter your present address:\n");

    printf("Name: ");
    scanf(" %[^\n]", a.name);

    printf("Home Address: ");
    scanf(" %[^\n]", a.home_address);

    printf("City: ");
    scanf(" %[^\n]", a.city);

    printf("State: ");
    scanf(" %[^\n]", a.state);

    printf("ZIP Code: ");
    scanf(" %[^\n]", a.zip);

    printf("\nYour Present Address is:\n");
    printf("%s\n", a.name);
    printf("%s\n", a.home_address);
    printf("%s, %s - %s\n", a.city, a.state, a.zip);

    return 0;
}
```

```
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output> & .\'exp4.exe'
Enter your present address:
Name: Sanchita
Home Address: H.NO. 121 New Jain Nagar
City: Yamunanagar
State: Haryana
ZIP Code: 135001

Your Present Address is:
135001
135001
135001, 135001 - 135001
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_6.c\output>
```